

■ SQL Reference Guide

Eleazar Soto — Data Analytics Portfolio

■ Level 1 — Basics

1. SELECT + FROM

```
SELECT columna1, columna2  
FROM tabla;
```

2. WHERE — Filter rows

```
SELECT columna1, columna2  
FROM tabla  
WHERE condicion;
```

3. IS NULL / IS NOT NULL

```
SELECT columna1 FROM tabla WHERE columna IS NULL;  
SELECT columna1 FROM tabla WHERE columna IS NOT NULL;
```

■ Level 2 — Aggregation

4. ORDER BY

```
SELECT columna1, columna2  
FROM tabla  
ORDER BY columna ASC; -- or DESC
```

5. COUNT / SUM / AVG / MAX / MIN

```
SELECT COUNT(*) AS total  
FROM tabla  
WHERE condicion;
```

6. GROUP BY + HAVING

```
SELECT columna, COUNT(*)  
FROM tabla  
GROUP BY columna  
HAVING COUNT(*) > 10  
ORDER BY COUNT(*) DESC;
```

7. LIKE — Pattern matching

```
WHERE columna LIKE 'valor%'; -- starts with  
WHERE columna LIKE '%valor%'; -- contains  
WHERE columna LIKE '%valor'; -- ends with
```

8. IN — Multiple values

```
SELECT columna1
FROM tabla
WHERE columna IN ('valor1', 'valor2', 'valor3');
```

9. BETWEEN — Range

```
SELECT columna1
FROM tabla
WHERE columna BETWEEN valor1 AND valor2;
```

■ Level 3 — Joins

10. INNER JOIN — 2 tables

```
SELECT t1.columna, t2.columna
FROM tabla1 t1
JOIN tabla2 t2 ON t1.columna_comun = t2.columna_comun;
```

11. INNER JOIN — 3 tables

```
SELECT t1.columna, t2.columna, t3.columna
FROM tabla1 t1
JOIN tabla2 t2 ON t1.columna_comun = t2.columna_comun
JOIN tabla3 t3 ON t2.columna_comun = t3.columna_comun;
```

12. LEFT JOIN

```
SELECT t1.columna, t2.columna
FROM tabla1 t1
LEFT JOIN tabla2 t2 ON t1.columna_comun = t2.columna_comun;
```

13. LEFT JOIN — Find unmatched records

```
SELECT t1.columna
FROM tabla1 t1
LEFT JOIN tabla2 t2 ON t1.columna_comun = t2.columna_comun
WHERE t2.columna IS NULL;
```

■ Level 4 — Advanced (In Progress)

14. CASE — Conditionals

```
SELECT columna,
CASE
    WHEN condicion1 THEN 'resultado1'
    WHEN condicion2 THEN 'resultado2'
    ELSE 'resultado_default'
END AS nombre_columna
FROM tabla;
```

15. CASE + JOIN + GROUP BY

```

SELECT t1.columna,
CASE
    WHEN COUNT(*) > 10 THEN 'VIP'
    WHEN COUNT(*) BETWEEN 5 AND 10 THEN 'Regular'
    ELSE 'New'
END AS categoria
FROM tabla1 t1
JOIN tabla2 t2 ON t1.columna_comun = t2.columna_comun
GROUP BY t1.columna;

```

■ Clause Order — Always follow this

```
SELECT → FROM → JOIN → WHERE → GROUP BY → HAVING → ORDER BY → LIMIT
```

■ Key Rules

Rule	Example
Text with single quotes	WHERE City = 'London'
Numbers without quotes	WHERE UnitPrice > 20
Filter groups	HAVING not WHERE
NULL comparison	IS NULL not = NULL
Unique alias per table	c, o, p, ca, s, sh, e
CASE uses real columns	WHEN UnitPrice < 10 not alias

■ Northwind — Table Connections

```

Customers ■■■■ Orders ■■■■ Shippers
        CustomerID        ShipVia = ShipperID

Products ■■■■ Categories        Products ■■■■ Suppliers
CategoryID        SupplierID

Employees ■■■■ Orders
EmployeeID

```